

streams, and a higher frequency band, along with wide bandwidths of up to 80 MHz. (The 160MHz bandwidth in this standard is achieved by using 80+80MHz mode.) Moreover, due to the 256 QAM, this standard requires an error vector magnitude (EVM) of

-32 dB. The R&S FSW-K91ac tool features a very low EVM of less than -45 dB.

Announced on 20th June this year from Munich, the FSW-K91ac option enables the R&S signal and spectrum analyser to record and demodulate the full bandwidth

of a WLAN signal in line with the new standard. It has a 31cm (12.1-inch) touchscreen with MultiView. MultiView provides users with simultaneous view of multiple measurements and applications. Pricing is available only on request.

2 Spectrum analysers for education and hobbyists

Spectrum analysers for educational use and hobbyists should not only let them experiment with measurement testing but also act as a standard industrial instrument.

The DSA800 series from Sciencetech includes compact, light-weight and low-cost spectrum analysers that suit the education and hobbyist applications. The digital IF technology of this series helps in radio frequency (RF) applications like measurement of an RF amplifier's characteristics, measurement of an RF bandpass filter's characteristics, and measurement of voltage standing wave ratio.

The analysers offer a minimum resolution bandwidth of 100 kHz and are available with or without

Key Features of Sciencetech's DSA800 Series

Features	Value
Low minimum resolution bandwidth (RBW)	100 kHz
Displayed average noise level	-135 dBm
Offset phase noise	-80 dBc/Hz @ 10 kHz
Total amplitude uncertainty	<1.5 dB
Interfaces available	LAN, USB host, USB device, GPIB

The price of the basic model starts from Rs 145,000 + tax with two-year warranty.

tracking generator. Their widescreen display, advanced measurement functions, electromagnetic interference (EMI) filter and quasi-peak detector kit, up to -135dBm displayed average noise level, phase noise of -80 dBc/Hz at 10kHz offset, total amplitude uncertainty of less than 1.5 dB, and the capability to inter-

face through LAN, USB host, USB device and GPIB make them suitable for a majority of education and hobbyist applications.

The series comprises spectrum analysers to be used in basic electronics, basis communication, antenna and wireless communication labs.

3 Millimetre-wave analysers

Millimetre-wave (mm-wave) analysers are intended for spectrum and signal analysis of emerging wideband communication systems. Using new capabilities, engineers can evaluate, characterise and manufacture products designed for emerging wideband standards, such as WiGig, including FCC Part 15 compliance emission testing requirements from 40 GHz to 200 GHz.

The MxxHWD harmonic mixer, based on a single-diode design, is available in waveguide bands from 26.5 GHz to 325 GHz. The harmonic mixer is a two-port frequency extension product with mm-wave interface

Key Features of Anritsu MS2830A

Features	Value
Frequency coverage	26.5 GHz to 325 GHz
Support for emerging wideband standards	WiGig, FCC Part 15 compliance emission testing
Dynamic range	Features innovative external mixer design
Low-level mm-wave measurement	Measures E-band (60-90GHz, WR-12) with better displayed average noise level (nominal -120 dBm/Hz @ 80 GHz)
Top RF specifications for displayed average noise level	-144 dBm/Hz @ 40 GHz
Third-order intermodulation distortion	+13 dBm @ 40 GHz

Pricing is available only on request. Meera Agencies is one of the authorised resellers for Anritsu products

for device-under-test (DUT) connection. The Anritsu MS2830A signal analyser, when coupled with the OML

MxxHWD harmonic mixer, offers mm-wave frequency coverage from 26.5 GHz to 325 GHz.